

OBJECTIVES

- Integration of a relational database within a mobile App.

INDIVIDUAL EXERCISES

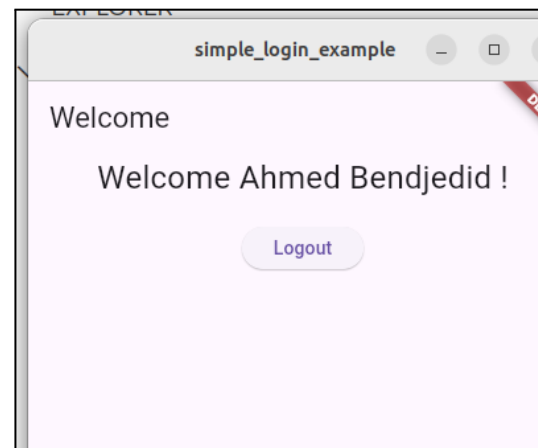
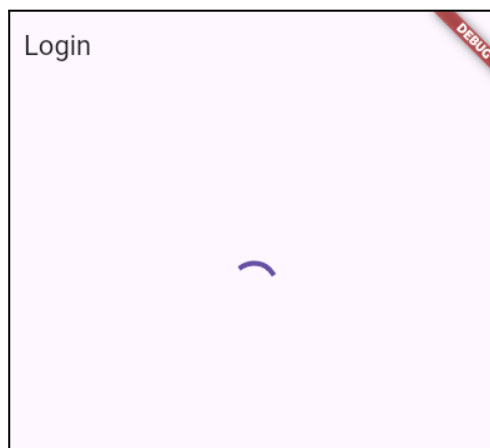
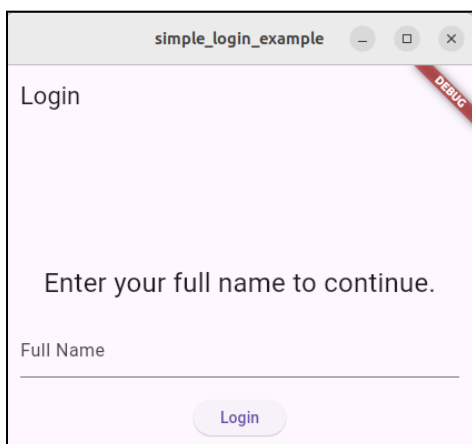
Task / Exercise C1 (SharedPreferences) [45 mins + 15 minutes demonstration]:

Create a simple form screen where the user enters their **full name**:

1. Simple Form validation check that the length of their full name is > 3 letters.
2. The user will be shown a circular progress bar for 2 seconds
3. The user will be redirected to the Welcome screen showing "Welcome student_full_name!" with a logout button

Use the SharedPreferences library for the following :

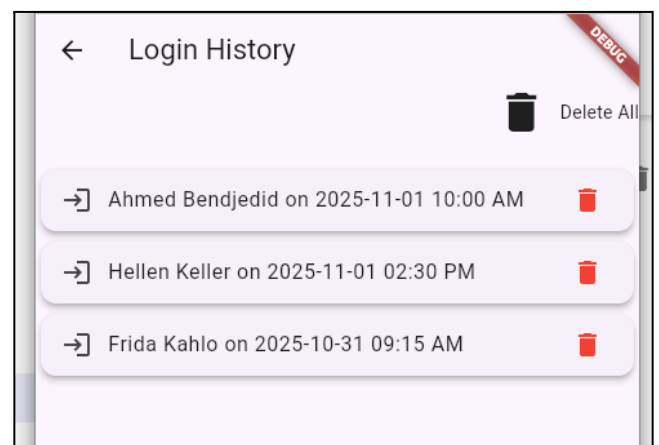
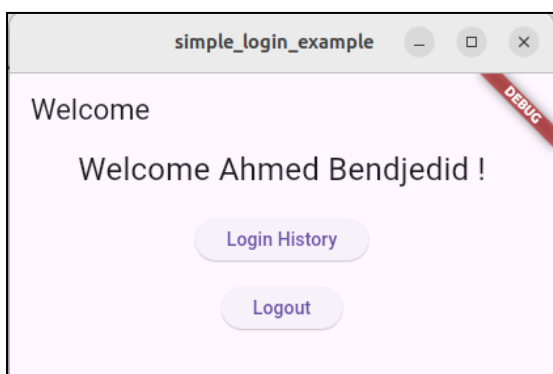
- If the user **closes** the app, and later opens the App, the user should be **automatically taken** to the welcome screen showing their name.
- If the user was already logged out, they will be shown the login screen.



Task / Exercise C2 (Login History using Relational Database) [60 mins+ 30 minutes demo]:

Extend C1 to show the recent login history being loaded from a local relational database when the user clicks on the button "Login History" as shown below.

- You need to use the Repository Design Pattern connected to an SQLite engine.
- You can copy the DB helper from the lecture slides.
- Implement the options for the user to : delete a single history line or Delete All



TEAM EXERCISES

Task / Exercise C3 (Team Project) [30 mins] :

Team members should progress on their projects to complete :

- All Screens
- Navigation between screens
- Create Dummy EndPoint functions when needed.
- Create the database tables

NEXT DELIVERABLE : In-Class Demonstration of the App.

Week 11 (30 Nov – 5 Dec), students will demonstrate an MVP with :

- The primary use cases are implemented.
- Minimal Backend
- Minimal Business Logic
- Use of State Management
- Local Relational Database.

NO OPTIONAL EXERCISES